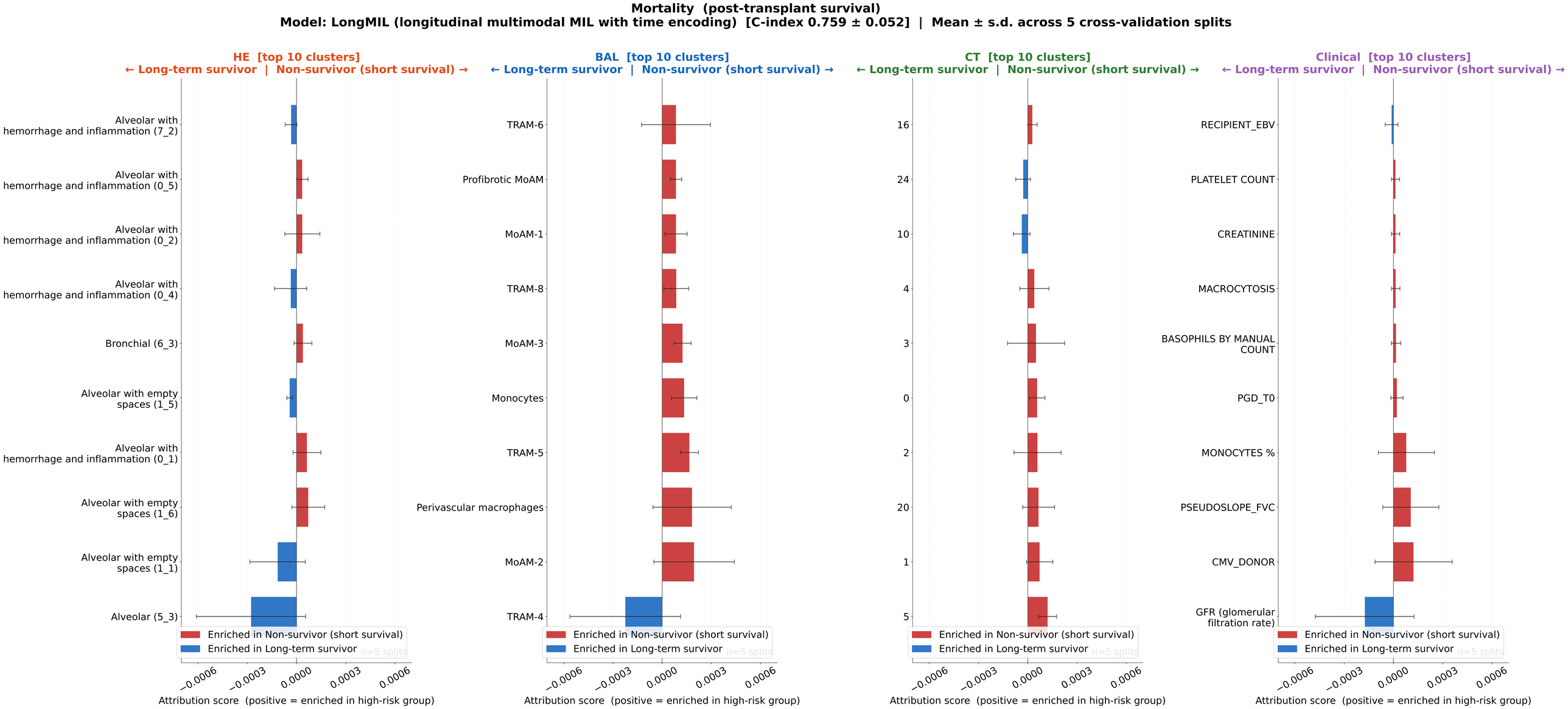
Mortality (post-transplant survival)
Model: LongMIL (longitudinal multimodal MIL with time encoding) [C-index 0.759 ± 0.052] | Mean ± s.d. across 5 cross-validation splits
HE [top 10 clusters]
← Long-term survivor | Non-survivor (short survival) →
Alveolar with hemorrhage and inflammation (7_2)
Alveolar with hemorrhage and inflammation (0_5)
Alveolar with hemorrhage and inflammation (0_2)
Alveolar with hemorrhage and inflammation (0_4)
Bronchial (6_3)
Alveolar with empty spaces (1_5)
Alveolar with hemorrhage and inflammation (0_1)
Alveolar with empty spaces (1_6)
Alveolar with empty spaces (1_1)
Alveolar (5_3)
Enriched in Non-survivor (short survival)
Enriched in Long-term survivor
n=5 splits
−0.0006
−0.0003
0.0000
0.0003
0.0006
Attribution score (positive = enriched in high-risk group)
BAL [top 10 clusters]
← Long-term survivor | Non-survivor (short survival) →
TRAM-6
Profibrotic MoAM
MoAM-1
TRAM-8
MoAM-3
Monocytes
TRAM-5
Perivascular macrophages
MoAM-2
TRAM-4
Enriched in Non-survivor (short survival)
Enriched in Long-term survivor
n=5 splits
−0.0006
−0.0003
0.0000
0.0003
0.0006
Attribution score (positive = enriched in high-risk group)
CT [top 10 clusters]
← Long-term survivor | Non-survivor (short survival) →
16
24
10
4
3
0
2
20
1
5
Enriched in Non-survivor (short survival)
Enriched in Long-term survivor
n=5 splits
−0.0006
−0.0003
0.0000
0.0003
0.0006
Attribution score (positive = enriched in high-risk group)
Clinical [top 10 clusters]
← Long-term survivor | Non-survivor (short survival) →
RECIPIENT_EBV
PLATELET COUNT
CREATININE
MACROCYTOSIS
BASOPHILS BY MANUAL COUNT
PGD_T0
MONOCYTES %
PSEUDOSLOPE_FVC
CMV_DONOR
GFR (glomerular filtration rate)
Enriched in Non-survivor (short survival)
Enriched in Long-term survivor
n=5 splits
−0.0006
−0.0003
0.0000
0.0003
0.0006
Attribution score (positive = enriched in high-risk group)